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Reflections eight years on from the first declaration of climate emergency: The role of LDA topic modelling combined with qualitative policy analysis in detecting a frame of "climate emergency" in real-world policy

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ABSTRACT

This paper considers the merits of combining LDA topic modeling as a technique in Natural Language Processing [NLP] and policy analysis to contribute to methods for systematically analysing the rapidly evolving climate policy landscape. The novelty of the use of topic modelling within our methods contributes to a growing literature on using NLP to analyse policy changes. In our case study, we consider the policy frame of "climate emergency". Eight years after the first declaration of climate emergency and now with the movement slowing, it is timely to reflect on the presence (or not) of the climate emergency policy mode. To do so, we undertake a text analysis of local government climate strategy documents of 70 local governments in Australia that declared a climate emergency from 2016 to the end of 2022. We aim to ascertain whether the framing of "climate emergency" can be detected in real-world policy utilising a mixed-methods approach of qualitative and quantitative methods, including the use of LDA topic modelling and qualitative policy analysis. We conclude that topic modelling techniques such as LDA contribute to the identification and analysis of the evolving framing of "climate emergency" in local governments' policies. LDA Topic modelling complements traditional qualitative policy analysis by introducing efficient and replicable methods for comprehensive examination of policy documents. In addition, for the first time at scale, we can assess the impact of the Climate Emergency Declaration movement across local governments in Australia, revealing the presence of all key attributes of the climate emergency mode.

1. Introduction

As all Parties to the United Nations Framework Convention on Climate Change (UNFCCC) turn their efforts toward implementation of the 2015 Paris Agreement, the role of non-party stakeholders (NPS, e.g., cities, states and regions, firms, and investors) becomes critical to deliver on net zero carbon by 2050. The number and membership of NPS continue to evolve, for instance, ranging from individual companies and sub-national governments to small groups of specific entities like the Global Covenant of Mayors for Climate and Energy, an alliance of over 10,000 cities and subnational entities, collectively representing nearly 10 per cent of the global population. The inclusion of NPS in the 2015 Paris Agreement marks a significant shift in global climate governance, now cementing multiple actors and institutions with considerable autonomy to set rules and norms in specific domains - a phenomenon which Ostrom (2010) referenced as 'polycentric governance'. This increasing complexity raises concerns about how to maintain policy coherence to advance collective climate objectives and targets (Chan et al., 2021; Jordan et al., 2018). Policymakers across government and business sectors recognise the drawbacks of a fragmented climate policy framework, which hinders effective coordination, accountability, and the implementation of unified climate measures.

Although an in-depth qualitative analysis of climate policies can help us understand the ambition in climate policy, the drastic rise in the number of climate actors and their policy documents hinder a clear and comprehensive view of the present situation. Qualitative case studies are generally limited to a small selection of cases, focusing on providing an 'in-depth and within its real-world contexts, especially when the

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boundaries between phenomenon and context may not be clearly evident' (Yin, 2009, p. 15). However, this method is limited in assessing the broader trends in climate governance, which understood at scale may well provide for a more substantive leverage to enable change (Wolfram, 2016). Natural Language Processing [NLP] and its methods, such as topic modelling, allow policy analysis to move beyond a select few cases, enabling systematic analysis of the rapidly evolving climate policy landscape (Hsu and Rauber, 2021). Automated large scale analysis potentially enables consistent evaluation between policies across different jurisdictions, track changes in real-time to timely identify emerging controversies and plays an important role in supporting policy development, supporting evidence-based data-driven decision-making.

To move forward on this limitation of qualitative policy analysis incorporating only a few cases, data science methods, mainly Natural Language Processing [NLP] models, are beginning to be utilised for quantitative and systematic analysis of large collections of policy inputs that are qualitative by nature. These models use statistical and machine learning methods to automatically analyse a large text corpus (Bohn and Roggee, 2022). Recently, topic modelling - one of NLP techniques - has been used to better understand the rapidly evolving policy landscape in this case climate policy. A prominent example is Hsu et al. (2021)'s study, which utilised topic modelling to understand the international landscape of climate actions taken by different climate actors (i.e., government and companies aligned with each other). However, topic modelling is more widely employed in the field of media studies, possibility due to the accessibility of data sources such as news articles and tweets. In media studies, topic modelling is utilised to understand the public discourse around specific climate topics. Bohn and Rogge (2022) draw upon Goffman definition of frames—in his seminal book about framing and frame analysis - as "schemata of interpretation" (Goffman, 1974, p. 21) to interrogate the public media framing of green innovations and electric vehicles. Dehler-Holland et al. (2021) utilised topic modelling to uncover shifts in media framing of the German *Renewable Energy Act*. Ylä-Anttila et al. (2022) undertook a comparative study of media debates on climate change in India and the United States. These studies clearly demonstrate that Latent Dirichlet Allocation [LDA] topic modelling can serve as a useful proxy for frames-in-frame analysis of public discourse debates. Despite increasing momentum, the application of topic modelling techniques remains largely under-explored in the fields of urban studies and climate governance. In particular, its capacity to track real-world climate policy frame, particularly when combined with qualitative analysis, has yet to be fully examined.

Here, we will consider the merits of topic modelling, and more specifically LDA technique, to contribute to methods for systematically analysing the rapidly evolving climate policy landscape, while combining it with deeper qualitative analysis. The novelty of the use of topic modelling within our methods contributes to a growing literature on using machine learning approaches to analyse policy changes. In our case study, we will consider the policy frame of "climate emergency" mode, understood as 'urgent mobilisation of economic and social resources at an abnormal level of intensity and scale '(Spratt, 2019: Davidson et al., 2020). Eight years after the first declaration of climate emergency and now with the movement slowing, i.e., 89 declarations in 2022/23 out of a total of 2346, it is timely to reflect on the presence (or not) of the climate emergency policy mode. To do so, we undertake a text analysis of local government climate strategy documents of 70 local governments in Australia that declared a climate emergency from 2016 to the end of 2022. We aim to ascertain whether the framing change of "climate emergency" can be detected in real-world policy utilising a mixed-methods approach of qualitative and quantitative methods, including the use of LDA topic modelling and qualitative policy analysis. Considering the fast-evolving climate policy landscape developing new ways to systematically analyse policy changes at scale is key to tracking the climate policy landscape, and here we argue the merits of using NLP techniques to analyse policy changes. We also extend the scope of previous studies on the climate emergency policy response with a focus on one or two detailed case studies (see Davidson et al., 2020; Salvia et al., 2023) by analysing a large corpus of local government climate policies over a eight-year period geographical scoped to Australia. Further, we extend Salvia et al.'s (2023) with their focus on specific indicators such as targets and whether plans combine both mitigation and adaptation actions, and here we consider whether a suite of attributes defined as the climate emergency mode (Davidson et al., 2020).

In addressing the identified research gaps, our key research questions are:

(1) How can topic modelling be combined with qualitative analysis to identify and analyze the presence of the framing of 'climate emergency' in local government climate policies?
(2) How has the declaration of a climate emergency by local governments in Australia influenced the adoption of the climate emergency mode into climate policies?

We structure this paper first by developing an understanding of emergencies as a climate policy response, and then we point to a framework that articulates the key attributes of the climate emergency mode (Davidson et al., 2020). Secondly, we detail the methodology for our research. Thirdly, we present our results from both DA topic modelling and qualitative policy analysis mapped within the ten key attributes of the climate emergency mode. Finally, we end with discussion and conclusion, and propose a research agenda moving forward.

2. The policy response of the climate emergency

Local governments have emerged as outspoken leaders instigating 'courageous conversations' about climate action in places where there has been a perceived absence of state and federal leadership (Davidson et al., 2020). In 2016, the Darebin City Council, in inner Melbourne, was the first jurisdiction in the world to declare a climate emergency. Subsequently, over 2325 local governments have made official declarations of climate emergency, confirming a trend towards local governments as a 'strategic front' (Jon, 2020; McGuirk et al., 2014). Generally, these declarations have three primary objectives: to advocate the need for climate action at higher levels of government, to galvanise communities to mobilise and to justify more urgent and targeted local policy responses (Davidson et al., 2020; Rode, 2019). In 2019, it was suggested that these declarations are among the strongest statements from government in the face of global warming (Smith-Schoenwalder, 2019). In 2020, the UN Secretary-General Antonio Guterres stated at a climate summit that 'world leaders should declare states of "climate emergency" in their countries to spur action to avoid catastrophic global warming' (ABC, 2020). Many climate emergency advocates argue that using the term 'emergency', when coupled with education and social organising, can elicit more urgent and appropriately scaled action (Klein Salamon, 2019; Rode, 2019) and harness hope that humans can still avert climate disaster.

In 2017, scientists signalled that the climate declaration movement could shift policy response from 'business as usual' towards a new emergency mode of climate governance (Sutton, 2017). Studies suggest that prior to the declaration of the climate emergency movement, local government policy officers conceptualised climate changes as a complex emergency but responded with 'business as usual' (Aldunce et al., 2016; Fünfgeld and McEvoy, 2014). However, the research remains inconclusive on whether a declaration of climate emergency results in accelerating climate action in local government climate plans (see Davidson et al., 2020; Nissen and Cretney, 2021; Salvia et al., 2023). Early in the movement, Davidson et al. (2020) highlighted the possibility of a shift from 'business as usual'; however, the dataset was limited to two local governments and first movers in developing climate emergency plans. Again in 2021, Nissen and Cretney (2021, p. 340) signalled the opposite and raised concerns 'that retrofitting an emergency approach to the climate crisis at the local scale has the potential to

reproduce status quo politics, and calls for a greater understanding of the diversity of approaches to emergency in climate politics'. A later study by Salvia et al. (2023) suggested that the declaration of climate emergency movement can motivate municipalities to become more ambitious on climate action, namely striving towards carbon neutrality and including objectives in plans on both mitigation and adaptation, with a focus on Italy.

Previous studies have primarily focused on detailed case studies utilising methods such as content analysis of individual local governments' climate plans such as Darebin, Melbourne and Auckland, New Zealand (Davidson et al., 2020), and/or interviews with key actors, such as in the cases of two local governments in New Zealand (Nissen and Cretney, 2021), and London's 33 municipalities (32 boroughs plus the City of London; Howarth et al., 2021). In addition, Salvia et al. (2023) focused on the case site of Italy involving 105 cities; however, their focus was on evaluating the presence of specific indicators such as targets and inclusion of both mitigation and adaptation in climate policies following the declaration of climate emergency rather than interpretation of a policy frame. Here, we will move beyond a small-scale selection of case studies and test the application of LDA topic modelling in combination with qualitative analysis. To undertake this real-word policy framing interpretation, we draw upon the conceptual framing of the climate emergency mode that incorporates 10 key attributes proposed by Davidson et al. (2020).

2.1. Key attributes of the climate emergency mode

Climate emergency mode means transcending 'business as usual' approaches, instead demanding radical, urgent mobilisation of economic and social resources at an abnormal level of intensity and scale (Spratt, 2019). Implicitly, this means recognising the complexity of climate change and implementing problem-appropriate complex responses. Davidson et al. (2020) proposed a framework identifying key dimensions of a problem-appropriate complex response (see Table 1). Their Climate Emergency Attributes framework (Davidson et al., 2020) builds on work by Fünfgeld and McEvoy (2014) and Aldunce et al. (2016); climate emergency advocacy by numerous key organisations such as Breakthrough and The Climate Mobilisation; and subsequent resources from Sutton (2017), Spratt (2019), Gilding (2019), Rode (2019), Klein Salamon (2019), Silk (2016) and Spratt (2019). The Climate Emergency Attributes framework includes ten key attributes. These attributes are necessarily broad, spanning governance and climate change-specific criteria. For a climate emergency response to recognise the complexity of climate change, these attributes cannot be considered mutually exclusive but rather interdependent and overlapping. By applying this framework to climate policy, we can assess the extent to which a more accelerated, transformational response, suitable for the complexity of the issue, is present.

3. Methodology

Our approach aims to determine the applicability of LDA, a topic modelling technique, in detecting the presence of real-world climate policy framing, particularly when combined with qualitative analysis.

To identify the policy topics, we developed a dataset that includes 93 local government climate policies in Australia from 70 local governments that Declared a Climate Emergency. These documents were initially collected via an automated search using the Beautifulsoup package in Python for all the local governments in Australia that have either declared or recognised climate emergency. If we did not find the policy through automated search, we searched for the respective local governments' policy manually. The climate policy landscape is complex, and a response to climate change can be found in various policy documents, e.g., a standalone climate policy, local strategic planning or sitting within a larger agenda of sustainability or environment. For our dataset, we included only policies that have a primary focus on climate

Table 1
Key Attributes of a Climate Emergency Mode.

N	Attribute	Description
1	Purpose of action	Does the plan truthfully state the 'what' and the 'why' of a climate emergency response? Does it plainly state who is responsible for action? Does it enable organisations as well as the communities they serve to mobilise behind a clear purpose of action (Spratt, 2019)?
2	Urgency of action	Does the plan necessitate rapid action (Gilding, 2019)?
3	Prioritisation of action	Have climate emergency actions have socio-governmental priority? Does the plan ensure that a climate emergency response is prioritised over policies incompatible with radical decarbonisation (Gilding, 2019; Spratt, 2019)?
4	Institutional resource mobilisation	Does the plan allocate available discretionary funds and other institutional resources such as technical capacity and technological assets towards action to ensure delivery of the objective (Gilding, 2016; Rode, 2019; Silk, 2016; Spratt, 2019)?
5	Social mobilisation	Does the plan actively empower the community to rally, support and work productively together to deliver climate action (Gilding, 2016; Klein Salamon, 2019)?
6	Restoring a safe climate	Does the plan include far-reaching mitigative efforts to restore a safe, decarbonised climate in conjunction with climate adaptation actions (occasionally within the resources referred to as 'building resilience' or 'reducing vulnerability') to rapidly address the causes of the climate emergency and lessen the impacts already being felt (Rode, 2019; Silk, 2016)? Does it encourage societal, economic, environmental and cultural transformations such as sustainable practices and patterns of production and consumption such as new technologies and corresponding changes in markets, user practices, and policy responses (Masson-Delmotte et al., 2018; Silk, 2016; Spratt, 2019)?
7	Adapting to a changing climate	Does the plan include adaptation efforts to restore a safe climate to rapidly address the causes of the climate emergency and lessen the impacts already being felt (Rode, 2019; Silk, 2016)?
8	Plan for informed action	Are the plan's targets, actions and monitoring based on current scientific data? Is there monitoring and evaluation capacity and research dedicated to closing knowledge gaps across varying aspects and developing critical solutions (Klein Salamon, 2019; Spratt and Dunlop, 2019)?
9	Coordination, partnerships and advocacy for action	Does the plan prescribe coordinated efforts between all sectors (Sutton, 2017)? Does it include advocating upward to state and national governments to support radical action? Building local capacity across council, their local communities and neighbouring local councils?
10	Equity and social justice	Given that the impacts of the climate emergency will have disproportionate impacts across society, does the plan ensure that both the burden of climate emergency action and the opportunities borne from a safe climate are equitably shared across local, national and even international communities (Spratt, 2019)?

change in their title, such as having the word 'climate' or one of the following key phrases related to climate action: 'net zero', 'emissions', 'energy', 'adaptation', 'renewable energy', 'climate resilience', 'cooling the city' and 'climate mitigation'. As 2016 marked the beginning of the climate emergency movement (with the first declaration), we selected relevant policies published between 2016 and 2022. However, in some cases, we included policies published before 2016 if no recent policy was available. Our automated search was undertaken to find urls that contained pdf files with the specified keywords on open-access websites. We then used Python to convert the pdf files to txt format and save them for further analysis, Most policies were published from 2019 (5 %), 2020 (37 %), 2021 (32 %) and 2022 (10 %). The average document size is 10,819 words, with a median of 8779 words. The smallest document contains 344 words, while the largest document contains 40,078 words.

Topic modelling is an unsupervised machine learning technique that assumes each word occurs in several topics, allowing for the capture of different meaning structures associated with a word. However, the word 'topic' in topic modelling does not refer to any specific algorithm feature. Instead, it interprets what the algorithm's output can be applied to, such as clusters formed by co-occurring objects. We argue that the outputs of topic modelling are best interpreted as traces of different ways of discussing a topic, referred to as frames (Bohn and Roggee, 2022; Ylä-Anttila et al., 2022). If all the text is related to the topic, the word co-occurrence patterns that emerge represent patterns of using certain words to talk about it, which are an approximation of policy framing patterns.

We used LDA, a topic modelling technique, to analyse the climate policies. LDA is a statistical method that measures the probabilistic distribution of a particular word throughout the corpus, aiming to uncover latent patterns by comparing the vocabulary of texts. These patterns are referred to as topics, which consist of clusters of frequently co-occurring words throughout the corpus. The analysis results in a number of topics, each described by a set of words of varying weights. Through this approach, we can analyse the importance of topics by examining the weight of each topic for a given document (Bohn and Roggee, 2022). To implement LDA on the climate policies, we used LDA Mallet implementation in Python to identify the topics within the climate policies.

We undertook the trial-and-error approach to identify the number of topics in the topic model because there is no definitive answer to how many topics the model should extract in advance. We concur with Carron-Arthur et al. (2016) that at some point, the model merges into more distinct topics (i.e., topics that are not too general and with little duplication in a low number of topics). Through trial and error, we settled on 30 topics (Table 2 provides an overview of the topics).

DiMaggio et al. (2013), Evans (2014) and Grimmer and Stewart (2013) have highlighted the importance of interpreting and validating topics when using topic modelling to analyse societal phenomena. In the present study, we propose that the word clusters generated by topic modelling can be interpreted as traces of frames. We utilise a two-fold qualitative analysis process to internally validate the topics and labels ascertained from the word clusters in the topic modelling.

Firstly, we implement the most widely applied internal validation method involving the analysis of representative text in the documents linked back to the word clusters (topics). In implementing this validation of topics, we focus on the prominence of the top three to four words of each word cluster to identify internally valid frames that link climate change to a coherent set of other concepts. We identified the policy documents as a part of our qualitative inspection by considering the total probability of each topic in one policy document. We utilise each topic's highest-scoring total probabilities in each policy document to identify key policies for the qualitative inspection. Each document distributes topics as probabilities, and the total probabilities of each topic in a document sum to 1.0. To ascertain the key policies associated with each identified topic, we initiated a selective analytical process. This involved isolating primary documents where the topic probability exceeded a predetermined threshold. Given the assumption of LDA that each

Table 2
Labels of each topic.

N	Topic label	Attribute of "climate emergency" mode
Topic 1	Identifying actions' impact area, external collaborators and enablers	Prioritisation of action; Coordination, partnerships and advocacy for action
Topic 2	Modelling scenarios to calculate and reduce emissions and move away from 'business as usual'	Purpose of action; Urgency of action
Topic 3	Reducing greenhouse gas (GHG) emissions, using LED streetlights and working with contractors	Restoring a safe climate
Topic 4	Local governments' actions based on master plans for reducing GHG emissions	Restoring a safe climate
Topic 5	Abatement of carbon dioxide emissions and the replacement of diesel	Restoring a safe climate
Topic 6	Council's functions and endorsement to support energy reduction with retrofit	Restoring a safe climate; Coordination, partnerships and advocacy for action
Topic 7	Climate crisis, highlighting sustainable mode of transport and learning from the pandemic	Restoring a safe climate; Coordination, partnerships and advocacy for action
Topic 8	Building alliances, collaboration with other councils and other levels of government and evaluation of policies	Plan for informed action
Topic 9	New employment hubs and affordable and clean energy	Equity and social justice
Topic 10	Innovation in energy use and delivery to prepare for carbon pricing	Restoring a safe climate
Topic 11	Cost and energy saving (per annum) and responsibilities, and waste management and timeframes	Restoring a safe climate
Topic 12	Promoting cross-team/cross-organisational collaboration to deliver the action plan	Coordination, partnerships and advocacy for action
Topic 13	Building resilience to disasters such as coastal erosion and bushfire	Adapting to a changing climate
Topic 14	Attention to manage bushfires	Adapting to a changing climate; Equity and social justice
Topic 15	Effectiveness of actions, carbon sequestration, adaptive actions	Restoring a safe climate; Adapting to a changing climate
Topic 16	Reducing emissions from farming and agriculture, protecting endangered animals	Restoring a safe climate; Adapting to a changing climate
Topic 17	Vertical policy alignment within the council and between the council and other governance levels	Coordination, partnerships and advocacy for action
Topic 18	Enabling community-led ideas and skills	Social mobilisation; Coordination, partnerships and advocacy for action
Topic 19	Roadmap for reducing the emissions to achieve carbon neutrality	Restoring a safe climate
Topic 20	Preparing for natural hazards and reducing the reliance on external sources of food and energy	Restoring a safe climate; Adapting to a changing climate
Topic 21	Urban cooling by greening the urban surfaces	Restoring a safe climate; Adapting to a changing climate
Topic 22	Adaptation to the seasonal changes in the climate	Adapting to a changing climate
Topic 23	Revegetating catchments and supporting sustainable agricultural practices	Restoring a safe climate; Adapting to a changing climate
Topic 24	Strategies for a thriving nature and its interconnection with humans	Restoring a safe climate; Adapting to a changing climate; Coordination, partnerships and advocacy for action; Equity and social justice
Topic 25	Drawing on the knowledge of Aboriginal Registered Parties to protect cultural sites from climate change impacts	Coordination, partnerships and advocacy for action; Equity and social justice

(*continued on next page*)

Table 2 (*continued*)

N	Topic label	Attribute of "climate emergency" mode
Topic 26	Attention to manage coastal hazards	Adapting to a changing climate
Topic 27	Encourage the public for their sustainability achievements through awards, determination to draw down carbon dioxide and eliminate emissions by encouraging divestment and retrofitting	Purpose of action; Urgency of action; Prioritisation of action; Restoring a safe climate
Topic 28	Eliminating plastic, greening the area, enabling affordable access to accredited renewable energy and preparedness for climate impacts	Restoring a safe climate; Equity and social justice
Topic 29	Generating and saving energy (kWh) from rooftop solar panels and the abatement of emissions	Restoring a safe climate; Coordination, partnerships and advocacy for action
Topic 30	Anticipating and approving budgets and resources and finding indicators for assessing the impacts of actions	Institutional resource mobilisation

document encapsulates 30 distinct topics, we established a baseline threshold at 0.03333 (equivalent to 1/30), reflecting the reciprocal value of the total number of topics. This benchmark serves as an indicator of average significance within the topic distribution of a document. While in theory a topic could assume a probability as high as 1.0 within a single document, such instances are extremely uncommon. Therefore, by applying the threshold of 1/30, we extracted key documents that attain a level of prominence at least equivalent to the average importance across the thirty topics. That is, we focus on the 'most dominant' corresponding to the topic number with the highest topic probability (see Appendix 1 that presents the two most dominant topics based on the distribution of topic probability in each document). On average, we inspected approximately two to three policy documents for each topic. For each policy document, we analysed the use of the 3–4 words across the document and extracted representative text from a policy (see Sections 4.1–4.7). At the end of this qualitative validation process, a draft label was attached to each topic which described the emphasis of the 3–4 predominant words in the word cluster (see Appendix 2 titled Topic label with identified dominant words and associated dominant policy documents (with their locations)). We also include Appendix 3 that includes additional representative quotes and authors' notes from policy documents that reflect 30 topic labels (in alignment with ten key attributes of the climate emergency mode) that is not included in the Results section. We include an Appendix 4 for completeness of analysis that outlines the word frequency of the words in CED documents for words above 150 frequencies in the vocabulary.

Secondly, we implemented an additional validation process whereby we analysed the stated goals and objectives of the dominant policy documents alongside our derived draft label for each topic. This qualitative cross-checking process considered whether the representative text analysis, topic labels and word cluster coincide to reflect what the policy writers stated as their intentions in the policy documents. Through this process, we made amendments as required to confirm the label of the word cluster. The results from the in-depth policy analysis in relation to identified topic labels are included in Appendix 5.

Once we had settled on labels for each topic, we then mapped these against the ten key attributes of the climate emergence mode. This mapping of labels against the attributes enabled us to detect a framing change of "climate emergency" in real-world policy change. To undertake this process, we utilised the outlined questions in Table 1.

Limitations of the study include we only test on topic method technique LDA and we acknowledge the limitations that include lack of frequency of use of a word, which sometimes translates to its lack of importance in the method of policy analysis. With the focus on dominant words, new emerging trends might not be identified. Further, between 2016 and 2022, many notable events may have been attributed to accelerated climate action alongside the declaration of climate emergency movement. For example, the release of two key international reports renewed warnings of the emergency: the IPCC 2018 report and the UN Environment Programme's 2019 Emissions Gap Report. We also note that 93 policy documents could be considered a small data set to utilise NLP, but we internally validate the topics utilising two methods, including the most common validation method of representative text but also cross-checking dominant policy documents stated overarching goals, objectives, and actions with the topic labels. This aligns with the aim of our research, which is to consider the role of NLP within a growing literature on using machine learning approaches to analyse policy changes.

4. Results

Our results are mapped to illuminate the presence, or lack thereof, of the climate emergency mode via the degree of engagement with the ten key attributes (see Table 1) in local government climate policies through 30 topics (see Table 2 and Table 3). In the sections below, we demonstrate how 30 topics appear in the ten key attributes of the climate emergency mode using examples of representative quotes from the policy documents (see Appendix 3 for additional representative texts and notes, and Appendix 4 for top dominant words identified in policy documents for each topic label).

Further, we manually cross-checked the stated goals of the climate policies to ascertain whether the representative text analysis and the topics settled on reflect what the policy writers stated as their intentions in the policy documents. Our purpose here is to incorporate one more qualitative cross-check process as a valuable way to gauge the internal and potential external validity of the method of utilising the outputs of LDA topic modelling, the word clusters, as traces of different ways of discussing a topic, referred to as frames. Through the results of the verification process, we assert the validity of our method, as the labels derived from the word clusters were consistent with the intentions of the policy documents. In particular, our in-depth policy analysis reveals a strong correlation (with no negative results) between the identified labels and the overarching goals, objectives, and actions outlined in the

Table 3
Summary of Number of topics in each key attribute.

N	Attribute	Total number of topics	Topic numbers
1	Purpose of action	2	2, 27
2	Urgency of action	2	2, 27
3	Prioritisation of action	2	1, 27
4	Institutional resource mobilisation	1	30
5	Social mobilisation	1	18
6	Restoring a safe climate	17	3,4,5,6,7,10,11,15,16,19,20,21,23,24,27,28
7	Adapting to a changing climate	10	13,14,15,16,20,21,22,23,24,26
8	Plan for informed action	1	8
9	Coordination, partnerships and advocacy for action	9	1,6,7,12,17,18,24,25,29
10	Equity and social justice	5	9,14,24,25,28

policy documents, reinforcing the robustness and reliability of our approach (see Appendix 4).

4.1. Attributes of purpose of action, urgency of action, and prioritisation of action

In this section, we group three attributes together due to their interrelationship between labels in the 30 topics.

Based on the results of topic modelling and policy analysis, *urgency of action* is represented by a policy response of moving away from 'business as usual' (Topic 2), as articulated by the City of Melbourne in 'Climate Change Mitigation Strategy to 2050: Melbourne Together for 1.5 °C'. Here, four scenarios are considered: 'business as usual', significant action to reduce emissions, accelerated action to reduce emissions and purchasing offsets from 2020. Scenarios two and three align with the Paris Agreement, aiming to hit net zero emissions by 2040. The plan expresses a preference to achieve 'the Significant Action Scenario and pursue actions for the Accelerated Action Scenario where there is the opportunity to go further', indicating a commitment to achieving this climate goal sooner than the original 2050 target (CoM 2020, p. 5). Additionally, the plan highlights the urgency for swift actions with *a clear purpose*, aiming to 'take bold action' within the caveat of 'limited powers and financial resources' (CoM 2020, p. 5). Such limited governance powers of local government clearly point to the complexity around responsibility, prompting the need for a collective response and the role of advocacy for policy change at the different levels of government.

Regarding *prioritisation of action,* we detect an ambition to deliver climate emergency actions with socio-governmental priority, as articulated in Topic 27. The Darebin 'Climate Emergency Plan 2017–2022' outlines the goal of restoring 'a safe climate at emergency speed by eliminating greenhouse [gas] emissions and enabling drawdown of excess carbon dioxide in the air' (p. 28). Here, the concept of drawdown goes beyond goes beyond carbon neutrality and acknowledges the need to address past emissions. Further, the plan emphasises divestment as a crucial element of prioritising action, defined as:

> 'a subset of ethical investment. It means removing investment from companies involved in the fossil fuel supply chain and the institutions that fund it. This can be most effective when linked with advocacy action urging the institution to stop investing in or funding fossil fuel projects'. (p. 79)

These statements clearly reflect the commitment to ensuring that the climate emergency response takes precedence over policies incompatible with radical decarbonisation.

4.2. Attribute of social mobilisation

Our analysis reveals that *social mobilisation* is not a dominant trend in local governments' climate response, only being featured in one topic – Topic 18. The Macedon Ranges Shire Council 'Cool Changes Woodend Region Climate Change Action Plan', outlines an objective to facilitate the development of community-led climate actions, stating that:

> 'Everyone in our community is welcome and encouraged to get involved and take ownership of this Plan. […] Also remember that this Plan captures the community thinking at a point in time. Ideas and actions will evolve over time as circumstances allow. If you have other ideas for projects and the willingness to kick-start them, we encourage you to do so. Together, we can improve the sustainability of the Woodend Region!' (p. 6).

4.3. Attribute of restoring a safe climate

Our results indicate the commitment of local governments to implement mitigative efforts aligned with *restoring a safe climate* while incorporating climate adaptation actions. The policy responses primarily focus on two key areas: employing technology to reduce emissions and promoting sustainability practices and patterns.

Regarding the former, we detect the prominent technology focus on LED lights (Topic 3), investigating hydrogen/electric replacement for gas/diesel sources (non-fleet) (Topic 5) and solar panels (Topic 29). Our policy analysis also suggests that such technologies are embedded within sustainability practices across building and urban planning, agriculture, energy, waste, and transport sectors.

Turning to the latter, we identify a wide range of sustainable practices and patterns of production and consumption incorporated within local climate policy responses, including master plans for reducing GHG emissions (Topic 4), supporting energy reduction with retrofit (topic 6), sustainable modes of transport (Topic 7); reducing emissions from farming and agriculture (Topic 16); reducing reliance on external food and energy sources (Topic 20); urban cooling (Topic 21); supporting sustainable agricultural practices (Topic 23); thriving nature and its interconnection with humans (Topic 24); divestment (Topic 27); eliminating plastic (Topic 28); and generating and saving energy (kWh) from rooftop solar panels the abatement of emissions (Topic 29). An example can be found in the City of Boroondara's 'Climate Action Plan', which outlines new technologies and policy responses for encouraging sustainable practices, such as electric-powered vehicles and bicycles (Topic 5), charging infrastructure, and support for local business hubs to reduce commuter travel (Topic 7).

4.4. Attribute of adapting to a changing climate

Data analysis suggests that local governments' climate policy responses reflect a strong focus on the impacts of climate change, particularly natural disasters such as bushfires (Topics 13 and 14), coastal erosion (Topic 13) and biodiversity threat (Topic 16). We also detect a range of adaptation efforts promoted within identified policy documents, including building resilience to disasters with an emphasis on preparedness (Topic 13), the importance of traditional land management practices to improve disaster resilience (Topic 13), building adaptive capacity (Topic 15), exploring climate vulnerabilities relating to a reliance on external forces and systems such as food and energy (Topic 20), urban cooling by greening the urban surfaces (Topic 21), tree planting in regional areas to adapt to seasonal changes in climate and corridors to protect wildlife (Topic 22), revegetating across landscapes with a focus on catchments (Topic 23), to nature-based infrastructure (Topic 24).

For example, the Byron Shire Council's 'Climate Change Adaptation Plan' acknowledges 'the severity and extent of the Black Summer Bushfire season […] attributed to a combination of effects brought on by a changing climate' (Topic 13) (p. 51). The plan also emphasises the importance of incorporating 'Aboriginal cultural heritage and traditional land management practices to improve disaster resilience' (p. 8). In addition, 'preparedness' is listed as a key sub-category of actions for adaptation actions alongside two other sub-categories of plans, policies, and physical controls.

4.5. Attribute of plan for informed action

Our topic modelling results show limited evidence reflecting the attribute of *plan for informed action.* Only Topic 8 (out of 30 topics) includes an emphasis on policy evaluation, as articulated by the City of Glen Eira in '*Our Climate Emergency Response Strategy 2021–2025*'. The strategy outlines an objective for reviewing and updating associated action plans annually to maintain currency and response to monitoring and evaluation findings, stating that:

> 'Wherever possible, the evaluation will draw on data available from reliable sources such as State and Federal Governments, peak bodies, universities and other research institutions. Glen Eira will also draw

on existing Council data and will establish some new measurement and monitoring systems to capture information necessary to fill any gaps'. (p. 37)

4.6. Attributes of coordination, partnerships and advocacy for action

Data analysis reveals strong evidence of the presence of attributes such as *collaboration, partnerships, and advocacy for action* in local climate policies, as illustrated in Topics 1, 6, 7, 17, 24, 25 and 29. Common terminologies used beyond 'collaborators' and 'partnerships' include 'alliances', 'advocates' and 'networks'.

We detect the ambition of fostering collaborations and partnerships across sectors and scales in numerous climate policies within our dataset. An example of cross-sectoral collaboration can be found in the Bass Coast 'Climate Change Action Plan 2020–2030', which clearly identifies collaborators for the plan implementation, including the council, community groups and social service organisations (Topic 1). For cross-scale collaboration, the City of Greater Geelong 'Climate Change Response Plan' outlines an objective for working with neighbouring councils and other regional-level organisations within the Barwon South West Climate Alliance to collectively address climate change threats (Topic 6).

4.7. Attribute of equity and social justice

Our results show the beginning of embedding components of *equity and social justice* in local climate policies, including an emphasis on the importance of access to affordable energy and technology (Topics 9 and 28), the importance of nature for humans to thrive (Topic 24), and the inclusion of First Nations knowledge to protect the country and culture from the impacts of climate change (Topic 25).

While identified policy documents demonstrate intent and commitment to support social inclusion, there exist challenges for effective implementation. For example, the Surf Coast Shire Council's 'Climate Emergency Corporate Response Plan 2021–2031' outlines an objective to 'collaborate and work together in partnership with our local Registered Aboriginal Parties to protect the country and culture from the impacts of climate change' (Topic 25) (p. 6). However, the plan reflects a lack of clear implementation mechanisms (i.e., funding and resources) to achieve the objective of increasing engagement with the First Nations people. In detail, most Indigenous-related actions in this plan are under the status of 'currently unfunded – subject to external funding or budget bid in future years' (p. 30).

4.8. Attribute of institutional resource mobilisation

Our results only show the emerging presence of *institutional resource mobilisation* in Topic 30. While it is difficult to illuminate evidence of reallocation of available discretionary funds to accelerate climate action, the dominant topic did emphasise anticipating and approving budgets and resources. We caution here, as the relevant policy document, City of Greater Dandenong's 'Climate Emergency Strategy 2020 – 2030 Action Plan' placed caveat on allocated funds still requiring approval in the Council's annual process.

4.9. Summary

Within our methods, we implement a double qualitative validation process to cross-check whether the draft topic labels (derived from word clusters generated via topic modelling) accurately reflect the policy goals and intents. In this section, we examine representative texts and in the Appendix 5 an analysis of the stated goals and objectives of the dominant policy documents alongside our derived label for each topic. Based on the outcomes of the verification process, we then mapped the 30 labels derived from the topic modelling to the 10 key attributes of the climate emergency mode. In Table 3 we map and tally the number of topics and key attributes of the climate emergency.

5. Detecting a framing change of "climate emergency" in real-world policy change

We set out to answer the research questions of firstly, "How can topic modelling be combined with qualitative analysis to identify and analyze the presence of the framing of 'climate emergency' in local government climate policies?" and secondly, "How has the declaration of a climate emergency by local governments in Australia influenced the adoption of the climate emergency mode into climate policies?"

In answering the former research question, we argue that our method that incorporates LDA topic modelling opens the possibilities of large-scale policy analysis to better understand the climate policy landscape of local government in Australia. For the first time, at scale we can assess the impact of the Climate Emergency Declaration movement across local governments in Australia. Our findings reveal a presence of all key attribute of the climate emergency mode, particularly with a strong focus on "Restoring a safe environment", "Adapting to a changing climate", and "Coordination, partnerships and advocacy for action". We could also reveal gaps in the attributes of relevance for the climate emergency mode at scale mainly "Institutional Resource Mobilisation", "Social mobilisation' and "Plan for informed action". A promising contribution made from this large-scale analysis is that it offers an insight into both the most and least dominant frames in climate policy, after eight years on from the first Declaration of Climate Emergency in 2016 in local government in Australia. In the absence of such analysis, existing policy discourses have been reliant on small number of case studies, which do not provide the broader understanding at scale needed to identify impact from this movement and gaps still present in the policy discourse. The insights offered by our analysis, therefore, are critical for informing decision-making and policy development, contributing to the creation of more effective and innovative climate mitigation and adaptation strategies at the local level.

Regarding the latter research question, our results show interesting patterns in the contemporary incorporation of climate emergency mode in local government policies. First, it is unsurprising that the attributes of "Restoring a safe environment" and "Adapting to a changing climate" have a strong presence in our identified topics. In the attribute "Restoring a safe environment", we identified growing maturity of the development of sustainable practice and patterns in the policy document (9 topics). The impacts of climate change, mainly disasters, were also dominant topics in "Adapting to a changing climate", which points to unique place-based challenges; for example, coastal councils seek to develop relevant climate strategies to address coastal erosion and other related hazards.

Second, we observe the growing trend where local governments are actively partnering and forging regional alliances with neighbouring councils to deliver collective responses to climate threats (9 topics reflecting the presence of "Coordination, partnership and advocacy for action" attribute). This finding implies that local governments are increasingly developing collaboration for change while addressing place-based (but interdependent) climate risks. Moloney and Fünfgeld (2015, p. 30) also note "local government climate change alliances constitute an emergent and effective form of adaptive and integrative governance" that can potentially overcome institutional, political, and structural obduracies to build adaptive capacity at local and regional scales. Regarding the latter, Westman and Castan Broto (2018, p. 212) suggest that climate governance is shifting towards "less hierarchical, more collaborative, and non-regulative steering arrangements" to foster innovation and experimentation. In this sense, we suggest that including social mobilisation in local climate actions is critical to increasing access to resources (e.g., information) and facilitating the introduction and adoption of new policy approaches and emission reduction technologies. Our analysis revealed social mobilisation did not feature as a dominant

topic, and therefore an area of improvement in policy development.

Third, we start to see efforts to embed equity and social justice in climate policy responses, but these attempts are, at best, incremental. This is mainly illustrated in the dominant topics focused on access to technology, such as electric charging stations. While these initial efforts are promising, they are insufficient to comprehensively address deeper socio-environmental injustices that exacerbate vulnerabilities in cities (Amorim-Maia et al., 2022). In this sense, it is critical for local governments to adopt intersectional approaches to tackling these inequalities in a more holistic manner, setting the foundation for more "concrete and integrated strategies of just urban adaptation and transformation" (Amorim-Maia et al., 2022, p. 1). Moreover, we observe a small presence of the attributes of "Urgency of Action", "Priorisation of action", "Social mobilisation", and "Institutional resource mobilisation". Emerging themes in "Urgency of action" and "Priorisation of action" including the emphasis on modelling scenarios to calculate and reduce emissions and move away from "business as usual"; draw down carbon dioxide and eliminate emissions by encouraging divestment and a more holistic way forward by incorporating Sustainable Development Goals. For social mobilisation, emerging themes are enabling community-led ideas and skill and incorporating First Nation knowledge.

6. Conclusion

The continued increase in net zero pledges and the development of independent climate policies represent a significant missed opportunity for deeper coordination between climate actors and a more ambitious climate change response. There is a need for clarity around the current policy landscape and how climate action is changing, particularly as we approach 1.5 °C in global warming. Through our text analysis of local government policies in Australia, we map dominant topics within the framework of attributes of climate emergency mode. Within our methods, we utilise topic modelling for the identification and analysis of the evolving framing of "climate emergency" in local governments' policies. We utilise LDA topic modelling and argue the outputs of word clusters are interpreted as traces of different ways of discussing a topic, referred to as frames.

We conclude that topic modelling techniques such as LDA contributes to the identification and analysis of the evolving framing of "climate emergency" in local governments' policies. Topic modelling complements traditional qualitative policy analysis by introducing efficient and replicable methods for comprehensive examination of policy documents. It also paves the way for continuous, real-time policy analysis at scale. An analysis at scale enables an understanding of patterns, possibility to track trends over time, and correlations. Here we can ascertain the presence of the climate emergency mode key attributes across a larger number of local governments in Australia. This enables us to better understanding the landscape of climate policy in local government in Australia on (lack of) climate action. The analysis points to areas of where policy development needs to improve such as equity and social justice, social mobilisation and institutional resource mobilisation.

While topic modelling is one of the techniques in NLP, NLP is rapidly growing with the rise of transformer models and Large Language Models (LLMs) such as Chatgpt, which have advanced capabilities in processing and generating text similar to human beings. There have been efforts to utilise LLMS to facilitate easier exploration of climate policies, for example at national level (https://www.climatepolicyradar.org/). However, there are concerns about the implications of using advanced NLP techniques in terms of validity and accuracy of the output and potential for generating biased content. Hence the landscape is rapidly evolving to customise LLMS to ensure validity of the information based on expert's advice, as in the case of ChatNetZero (https://chatnetzero.ai/). Our paper contributes to these efforts by highlighting the value of expert human analysis in improving and critically evaluating the results of NLP models, with a focus on topic modelling. Therefore, we believe our approach provides a promising avenue to incorporate expert framing on climate emergency to train LLMS, with mor extensive applications to new climate policies.

To build upon our research, we propose the following research agenda. Firstly, we can extend this study utilising LDA topic modelling to map dominant topics across other policy actors such as corporations to further our understanding on the landscape of climate policy. By more widely utilising LDA topic modelling in policy analysis will hopefully bring attention to the importance of policy analysis at scale to better understand points of progression and controversies in the climate policy landscape. Secondly, for this current study while we have mapped the dominant topics against the attributes of climate emergency mode, the next step would be to undertake a verification process with key climate policy experts utilising direct interviews and focus groups. This process would also enable us to further our insights into the 'how' and 'why' these topics are dominant over other possible topics, identify missing topics and also develop an understanding of any emerging topics (that are absent from our process as we only detect dominant topics). Thirdly, future research can undertake an analysis of the location of the local governments featured in the analysis of the dominant topics. This will start to develop an understanding of which local governments are leading in action, why and if their location (urban or regional), and socio-economics have implications for accelerating climate action.

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CRediT authorship contribution statement

Sang Zichao: Methodology, Formal analysis, Data curation. **Mokhles Sombol:** Methodology, Formal analysis, Data curation, Writing – review & editing. **Nyuyen Thi Minh Phuong:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Davidson Kathryn:** Writing – review & editing, Writing – original draft, Validation, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

The authors have no relevant financial or non-financial interests to disclose.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.envsci.2025.104035.

Data availability

Supplementary material provided

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